

Characteristics of Forces — Paper D

HARD • 50 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- Every answer must carry a **unit**, and every resultant force must carry a **direction**. An answer with no unit does not score the final mark.
- The mark for each question is shown in square brackets on the right.
- No formula is given on this paper. Values of g are given only where they are not Earth's.
- Several questions describe situations you may not have seen before — apply the ideas you already have.

SECTION 1 – SEVERAL FORCES AT ONCE

- 1.** A crate of mass 40 kg is pulled along a rough floor by a horizontal rope. The rope pulls with 260 N and friction between the crate and the floor is 100 N.

(a) Calculate the weight of the crate. [1]

(b) State the size and direction of the force the floor exerts on the crate, and explain how you know. [2]

(c) Calculate the horizontal resultant force and describe the motion of the crate. [2]

- 2.** Two children pull a sledge along the ice in opposite directions. Priya pulls east with 145 N and Kabir pulls west with 190 N. Friction between the sledge and the ice is 15 N.

- (a)** Calculate the resultant of the two pulls, with its direction. [2]

- (b)** State the direction of the friction force, and calculate the overall resultant force on the sledge. [2]

- (c)** Describe the motion of the sledge. [1]

SECTION 2 – REASONING ABOUT MOTION

- 3.** A lift carrying a passenger of mass 60 kg is moving upwards at a constant speed.

- (a)** Calculate the weight of the passenger. [1]

- (b)** State the size of the upward force that the floor of the lift exerts on the passenger, and explain your answer fully. [3]

- 4.** A parachutist has a weight of 750 N. The moment her parachute opens, the air resistance acting on her rises to 1100 N.

- (a)** Calculate the resultant force on her and give its direction. [2]

- (b)** She is still moving downwards at that moment. Explain how she can be moving downwards while the resultant force on her acts upwards. [2]

SECTION 3 – WORKING BACKWARDS, AND DIAGNOSIS

- 5.** On the planet Zeta, an object of mass 60 kg has a weight of 138 N.

- (a)** Calculate the gravitational field strength on Zeta. [2]

- (b)** State what the same object would weigh on Earth. [1]

- (c)** State the mass of the object on Zeta. [1]

- 6.** A student draws a force diagram for a car driving along a straight, flat road at a steady 20 m/s. He draws one arrow only: a long arrow pointing forwards, labelled “driving force”.

- (a)** Name two forces he has left out, and state the direction of each. [2]

- (b)** Explain what is wrong with drawing the forward arrow longer than any backward arrow for this particular car. [2]

7. A student has written two answers, and both are wrong. [4]
- (i)** "The box has a mass of 20 kg, so its weight is 20 kg."
- (ii)** "The trolley is moving at a steady speed, so there must be a resultant force pushing it forwards."
- For each one, explain the mistake and give the correct statement.

End of paper. Check every answer has a unit or a form the question actually asked for.